Local-first Coordination via Friend-to-Friend Choreographies

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Pisa, 18-19 November 2026

– Motivations –

A framework for

[1] See [4, 5, 2, 1] for a formal model of local-first coordination

A framework for

friend-to-friend [7, 8]

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local-first [3, 6] coordination[1]

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A framework for

friend-to-friend [7, 8]

local-first [3, 6] coordination[1]

and its (forthcoming) implementation

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Blessing and curse of clouds

The cloud, our friend!

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cloud computing = fast communication infrastructures + global-scale computation facilities

- anytime-anywhere motto
- SaaS motto
- enabling dynamic reconfiguration (replace services if needed)
- reduced costs for maintenance, personnel, software, hardware, $\cdots$

Except when it is not



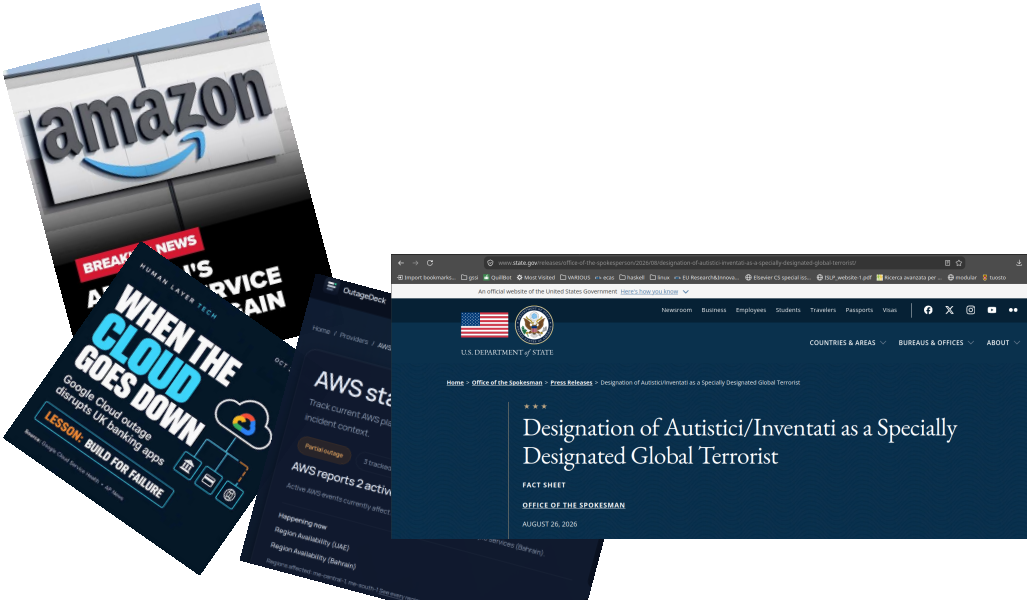
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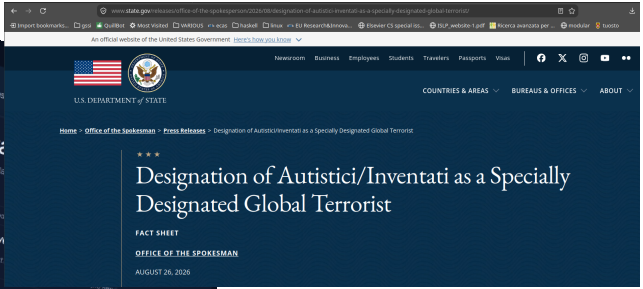


Except when it is not



So, what about going *local-first* ?

- minimise reliance on the cloud
- retain control over
  - data (i.e., ensure *confidentiality*)
  - computational resources
- have more "autonomy"



What if we went friend-to-friend too?

In Li & Dabek's words [7]

F2F: reliable storage in open networks

Abstract [...] We argue that a node should choose its neighbors (the nodes with which it shares resources) based on existing social relationships instead of randomly. This approach provides incentives for nodes to cooperate and results in a more stable system which, in turn, reduces the cost of maintaining data. [...]

In Popescu, Crispo & Tanenbaum words [8]

[...]Friends Team-Up and Beat the System

Abstract [...] while existing peer-to-peer architectures with similar aims attempt to build trust relationships on top of the basic, trust-agnostic, peer-to-peer overlay, Turtle takes the opposite approach, and builds its overlay on top of pre-existent trust relationships among its users. [...]

– A new choreographic model –

Our choreographic model

Syntax

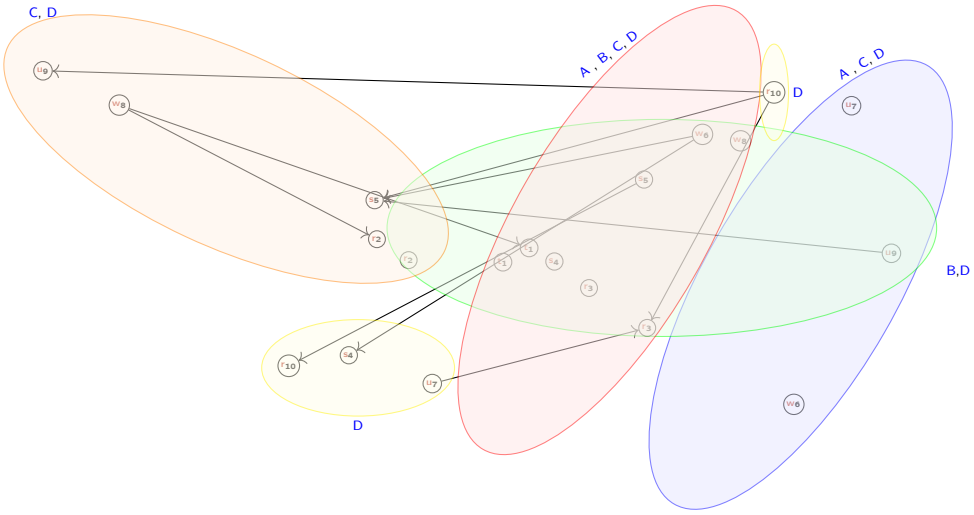
$$
\begin{aligned}
F \;::=\; & \mathsf{A} \rightsquigarrow \mathcal{B} : \mathsf{t} \\
\mid\; & F + F' \\
\mid\; & F ; F' \\
\mid\; & F \mid F' \\
\mid\; & \cdots
\end{aligned}
$$

Semantics

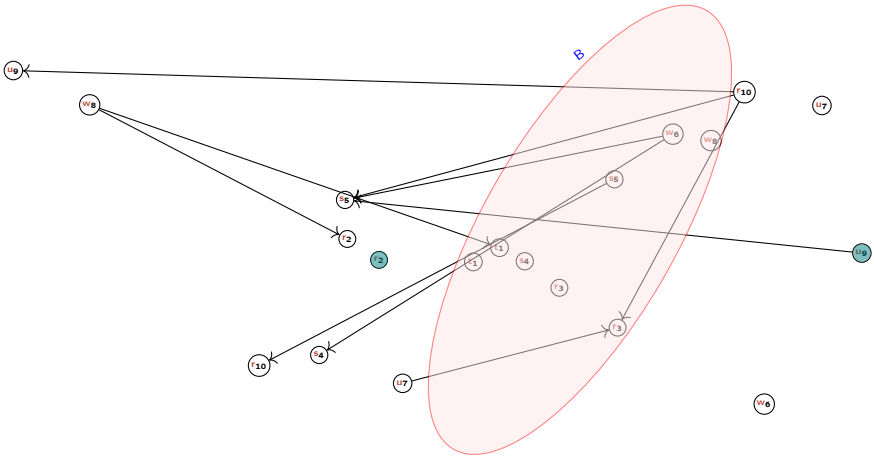
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\begin{aligned}
[\![\mathsf{A} \rightsquigarrow \mathcal{B} : \mathsf{t}]\!] &= \left\{ \left[\mathsf{A}!\mathcal{B}\langle \mathsf{t} \rangle \right] \right\} \\
[\![F + F']\!] &= [\![F]\!] \cup [\![F']\!] \\
[\![F ; F']\!] &= \{ r ;_{\mathcal{E}_r} r' \mid \cdots \} \\
[\![F \mid F']\!] &= \{ r ;_{\emptyset} r' \mid \cdots \} \\
[\![\cdots]\!] &= \cdots
\end{aligned}
$$

where
- $\mathcal{B}$ is a finite (possibly empty) set of A's "buddies"
- t is a "type"

The coordination model, informally



Replication

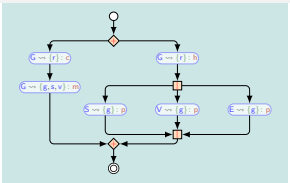


An F2F-choreography for fun

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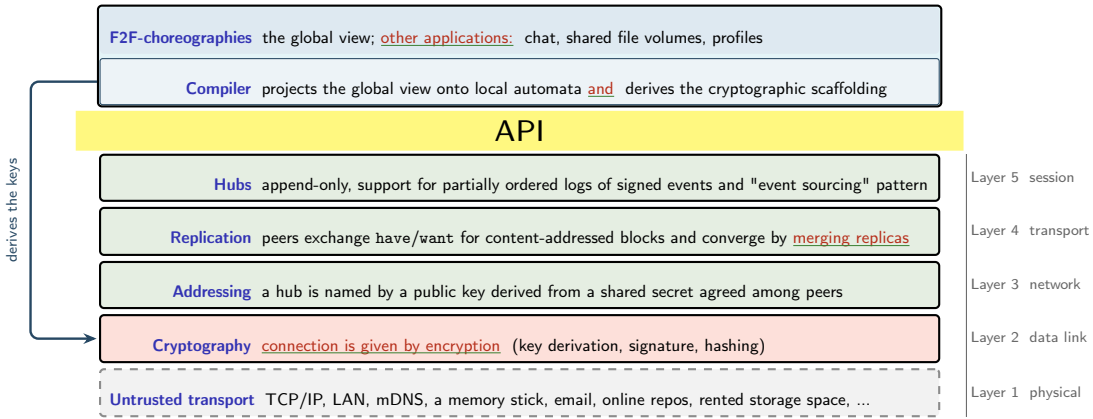


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[\![F]\!] \quad = \quad \left\{ \left[\begin{array}{c} \mathsf{G}!r\langle \mathsf{c} \rangle \\ \downarrow \\ \mathsf{R}!\{\mathsf{G},\mathsf{S},\mathsf{V}\}\langle \mathsf{m} \rangle \end{array} \right], \left[\begin{array}{ccc} & \mathsf{G}!r\langle \mathsf{h} \rangle & \\ \swarrow & \downarrow & \searrow \\ \mathsf{V}!\mathsf{G}\langle \mathsf{p} \rangle & \mathsf{S}!\mathsf{G}\langle \mathsf{p} \rangle & \mathsf{E}!\mathsf{G}\langle \mathsf{p} \rangle \end{array} \right] \right\}
$$

F2F-choreographies on top of Nearbytes



Hubs' addressing (hush-hush...no key distribution)

Buddies agree on a "master secret" M
Leverage buddies' public keys

$$\begin{cases} H & = M \cdot A^+ \cdot B^+ \cdot C^+ \cdot D^+ \cdot E^+ \\ H_1 & = M \cdot A^+ \cdot B^+ \cdot C^+ \cdot D^+ \\ \dots & \\ H_{31} & = M \cdot A^+ \\ H_{32} & = M \end{cases}$$

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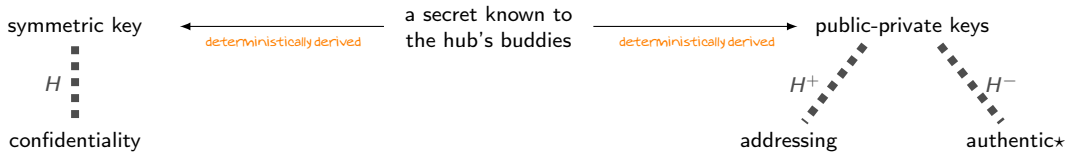
$$\text{payload} = \text{block}_m, \quad \text{block}_t, \quad \text{event}$$

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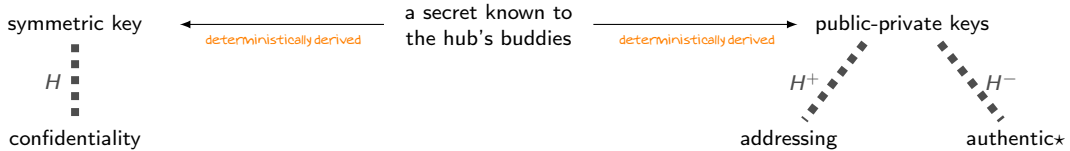


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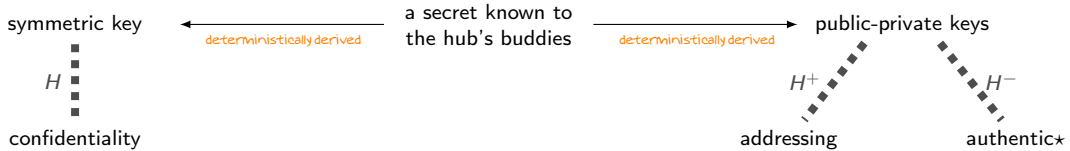
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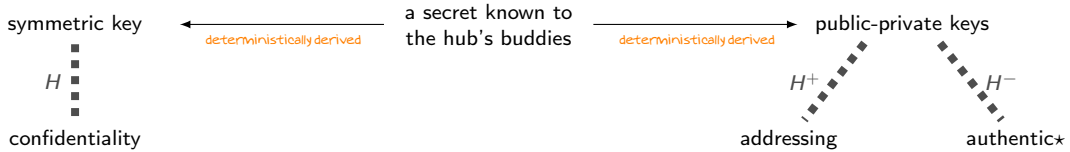
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Projections

Deriving buddies' expected behaviour:

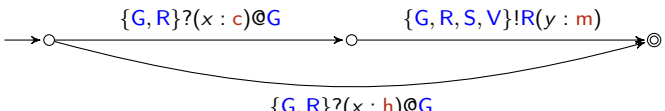
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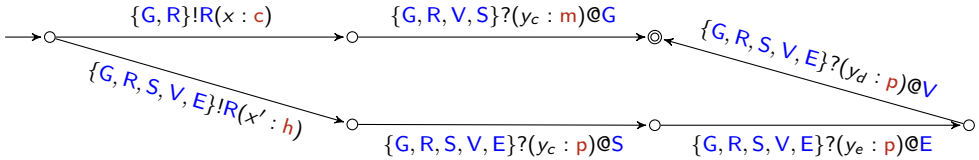


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$F|_{\mathsf{G}} =$



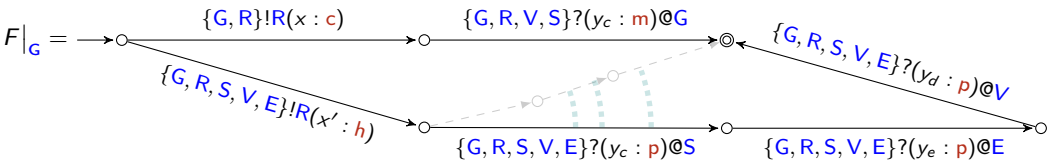
$$\left[\; \mathsf{G}!\{\mathsf{G}, \mathsf{R}, \mathsf{S}, \mathsf{V}, \mathsf{E}\}\langle \mathsf{h} \rangle \;\right]$$

$e = \{\, \mathsf{G}\text{'s address} \,\}_{H}, \{\, \mathsf{h} \,\}_{H^+}, \{\, \mathsf{hash}(\{\, \mathsf{G}\text{'s address} \,\}_{H}), \mathsf{hash}(\{\, \mathsf{h} \,\}_{H^+}) \,\}_{H^-}$

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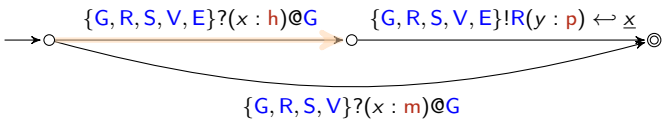
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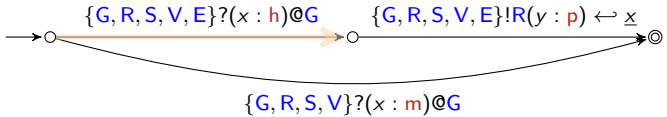
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$$\{\,\mathit{margherita}\,\}_H, \{\,\mathsf{p}\,\}_{H^+}, \{\,\mathsf{hash}(\mathsf{e}), \mathsf{hash}(\{\,\mathit{margherita}\,\}_H), \mathsf{hash}(\{\,\mathsf{p}\,\}_{H^+})\,\}_{H^-}$$

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- Compile the automata to actual multithreaded code (e.g., `Typescript` or `GoLang`) and use Nearbytes as a library

– Epilogue –

"Obscured by clouds"?

Bridges burning gladly
Merging with the shadows,
Flickering between the lines.
Stolen moments floating softly on the air,
Borne on wings of fire
and climbing higher.

Ancient bonds are breaking,
Moving on and changing sides.
Dreaming of a new day,
Cast aside the other way.

Magic visions stirring,
Kindled by and burning flames rise in her eyes.

The doorway stands ajar,
The walls that once were high.
Beyond the gilded cage,
Beyond the reach of ties.
The moment is at hand.

She breaks the golden band. (Pink Floyd)

Future work

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- Study well-formedness conditions of F2F-choreographies for "odd" protocols like

$$\mathsf{A} \rightsquigarrow \mathsf{m}_1; \mathsf{A} \rightsquigarrow \mathsf{m}_2; \mathsf{B} \rightsquigarrow \mathsf{m}_3 + \mathsf{A} \rightsquigarrow \mathsf{m}_2; \mathsf{B} \rightsquigarrow \mathsf{m}_4$$

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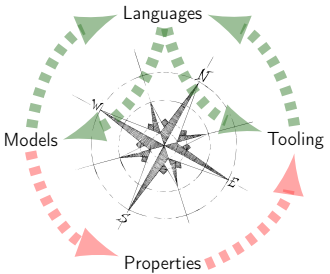
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  - Broadcast encryption schemes?
  - Use higher-dimensional automata?

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As Giangi taught us!

Thank Giangi!

References I

[1] Lucas Clorius, Florian Furbach, Roland Kuhn, Hernán Melgratti, Alceste Scalas, and Emilio Tuosto. Compositional Design, Implementation, and Verification of Swarms (Artifact).
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